

Program 5: CI Integration using Jenkins in Docker

Step 1: Check prerequisites & create project folder

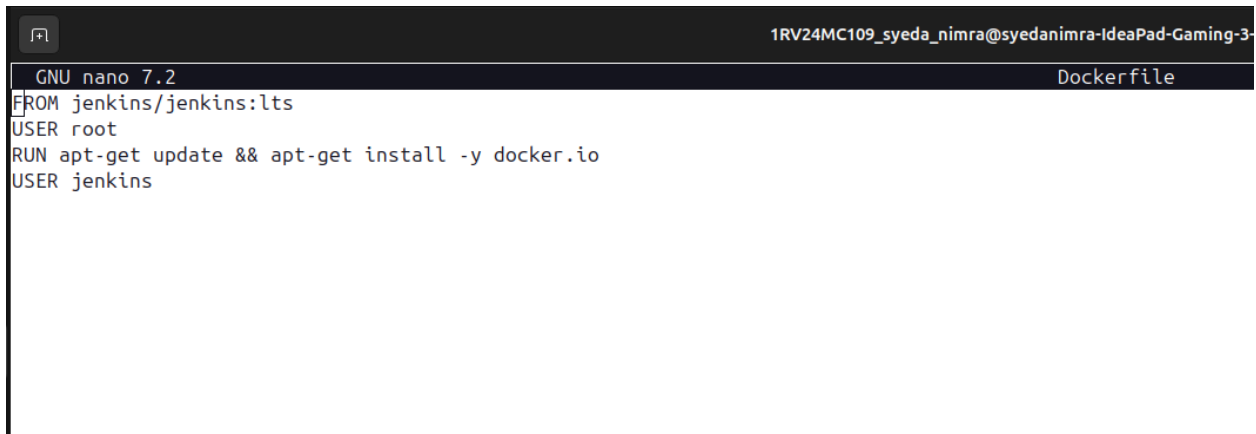
```
docker --version
docker compose version
mkdir program5
cd program5
```

Step 2: Create Dockerfile

nano Dockerfile

```
FROM jenkins/jenkins:lts
USER root
RUN apt-get update && apt-get install -y docker.io
USER jenkins
```

Save & exit.



```
1RV24MC109_syeda_nimra@syedaninimra-IdeaPad-Gaming-3
GNU nano 7.2 Dockerfile
FROM jenkins/jenkins:lts
USER root
RUN apt-get update && apt-get install -y docker.io
USER jenkins
```

Step 3: Create docker-compose.yml

```
nano docker-compose.yml
version: '3.8'
```

```
services:
  jenkins:
    build: .
```

container_name: jenkins-ci

ports:

- "8080:8080"
- "50000:50000"

volumes:

- jenkins_home:/var/jenkins_home
- /var/run/docker.sock:/var/run/docker.sock

volumes:

jenkins_home:

```
1RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05: ~/program5
GNU nano 7.2 docker-compose.yml
version: '3.8'
services:
jenkins: build: •
container_name: jenkins-ci ports:
- "8080: 8080"
- "50000:50000"
volumes:
- jenkins_home: /var/jenkins_home
- /var/run/docker.sock:/ var/run/docker.sock
volumes:
jenkins_home:
```

Step 4: Build and run Jenkins

docker compose up --build

```
1RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05: ~/program5
1RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/program5$ docker compose up --build
WARN[0000] /home/1RV24MC109_syeda_nimra/program5/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid
[+] Building 198.3s (8/8) FINISHED
=> [internal] load local bake definitions
=> => reading from stdin 533B
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 136B
=> [internal] load metadata for docker.io/jenkins/jenkins:ls
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/2] FROM docker.io/jenkins/jenkins:ls@sha256:1e407dfa9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7
=> => resolve docker.io/jenkins/jenkins:ls@sha256:1e407dfa9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7
=> => sha256:1e407dfa9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 3.12kB / 3.12kB
=> => sha256:75109de20007f9345278f298f4534d443409e234ba81225815caa812cc605644 12.71kB / 12.71kB
=> => sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc 65.08MB / 65.08MB
=> => sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471 5.24MB / 5.24MB
=> => sha256:b768bb699b5f1b6579e23e7253ae1157a339a20a2e9befdf8e03923e893b800 2.58kB / 2.58kB
=> => sha256:13cc39f8244ac66bf1dd9149e1da421ab1bbc80d612dc14fe368753e7be17b33 49.29MB / 49.29MB
=> => sha256:68cdef48009293f6a12f5c50b91fcfe4194bd629334d8a31a45b1ed63cb91a5 1.23kB / 1.23kB
=> => sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31168 183B / 183B
=> => sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d80688c35e7f1ed8627 95.32MB / 95.32MB
=> => sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaac0b98 189B / 189B
=> => extracting sha256:13cc39f8244ac66bf1dd9149e1da421ab1bbc80d612dc14fe368753e7be17b33
=> => sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2fd88e9c009f6874e7221acb06245 6.29MB / 6.29MB
=> => extracting sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc
=> => sha256:d7dd2e63023897146ec4b913a460382968fbd1cc3682e36f650727a1bfd6899 72.05MB / 72.05MB
=> => extracting sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471
=> => extracting sha256:68cdef48009293f6a12f5c50b91fcfe4194bd629334d8a31a45b1ed63cb91a5
=> => extracting sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31168
```

Step 5: Access Jenkins UI

Open browser:

<http://localhost:8080>

